



Neuro-fuzzy control & applications

Fuzzy Torque Vectoring control for Formula Student Electric Race car

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Introduction to Formula Student



P23 on the FSG Endurance track

General Concept:

- Formula Student is a student engineering competition held annually
- Teams from around the world design, build, test, and race a small-scale formula style racing car
- Vehicles are timed and a score is awarded based on the fastest vehicle to complete an event

Dynamic Events:

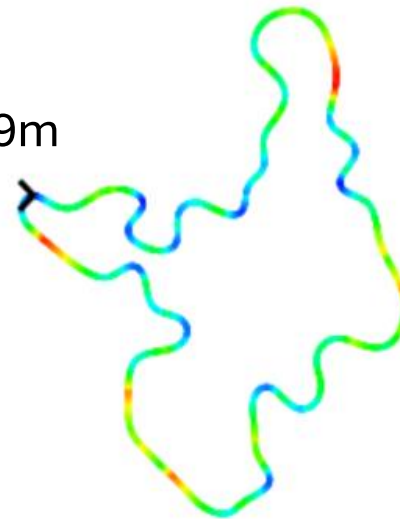
- Skid pad
- Autocross
- Endurance
- Acceleration

Dynamic Event Track Layout

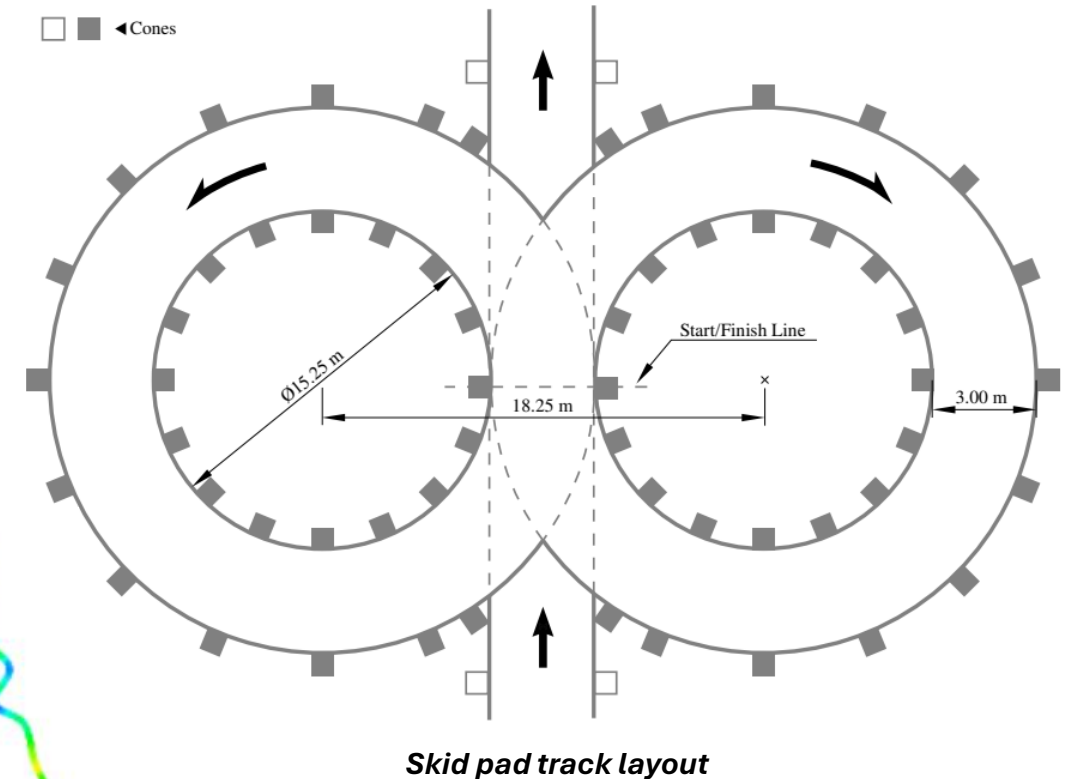
- Narrow tracks, consisting of multiple turns with short straights
- Quick car rotation and reliable car control is key to quickly complete laps and maximize scoring
- Utilizing independently driven wheels, teams can significantly enhance the vehicle's ability to rotate

Track Specifications:

- Minimum outside diameter of turns: 9m
- Minimum track width: 3m
- Minimum slalom distance: 7.5m
- Maximum straight length: 80m



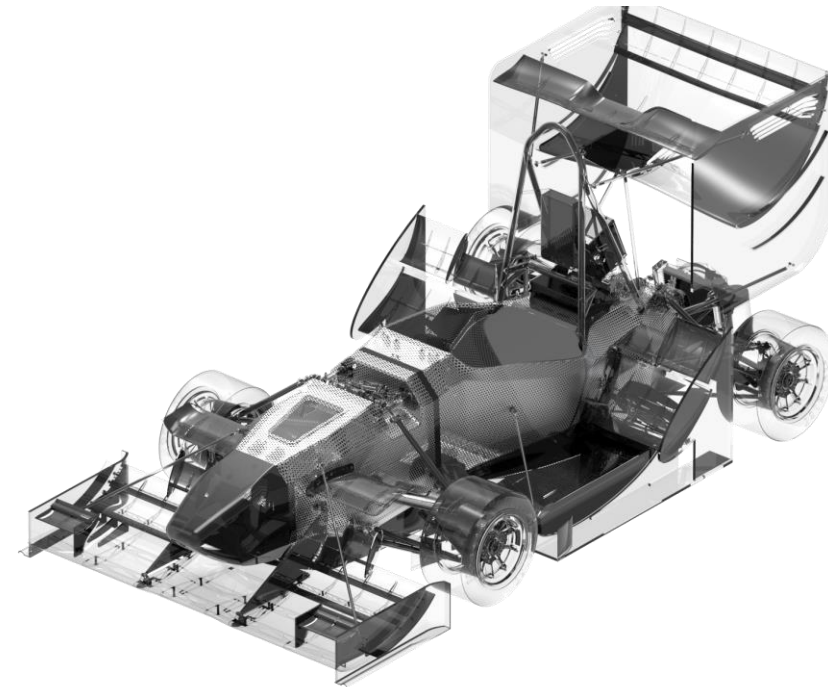
FSG Endurance track layout



Skid pad track layout

Car Specifications

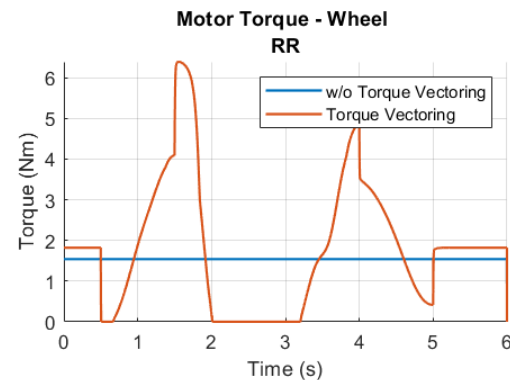
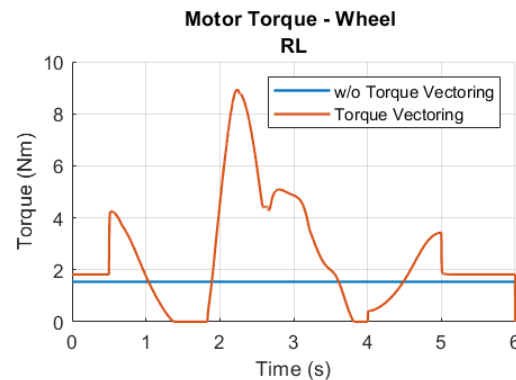
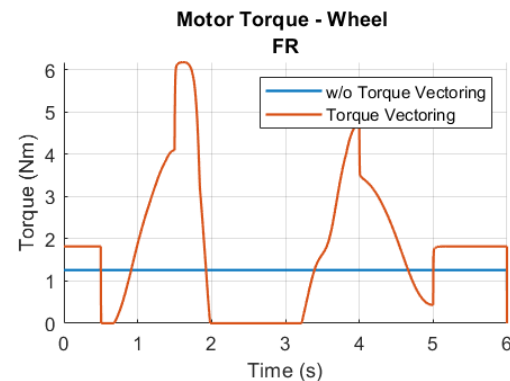
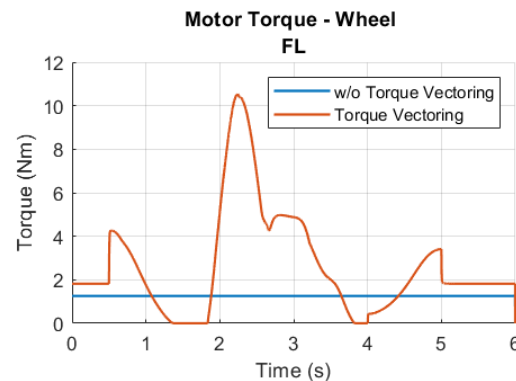
Spec	Value
Mass (Driver incl.)	270 [kg]
Weight Distribution	50%
Wheelbase	1570 [mm]
Vehicle Track	1220 [mm]
CoG Height	300 [mm]
Yaw Inertia	100 [kgm ²]
Tire Radius	203 [mm]
Gear Ratio	15



P23 CAD drawing

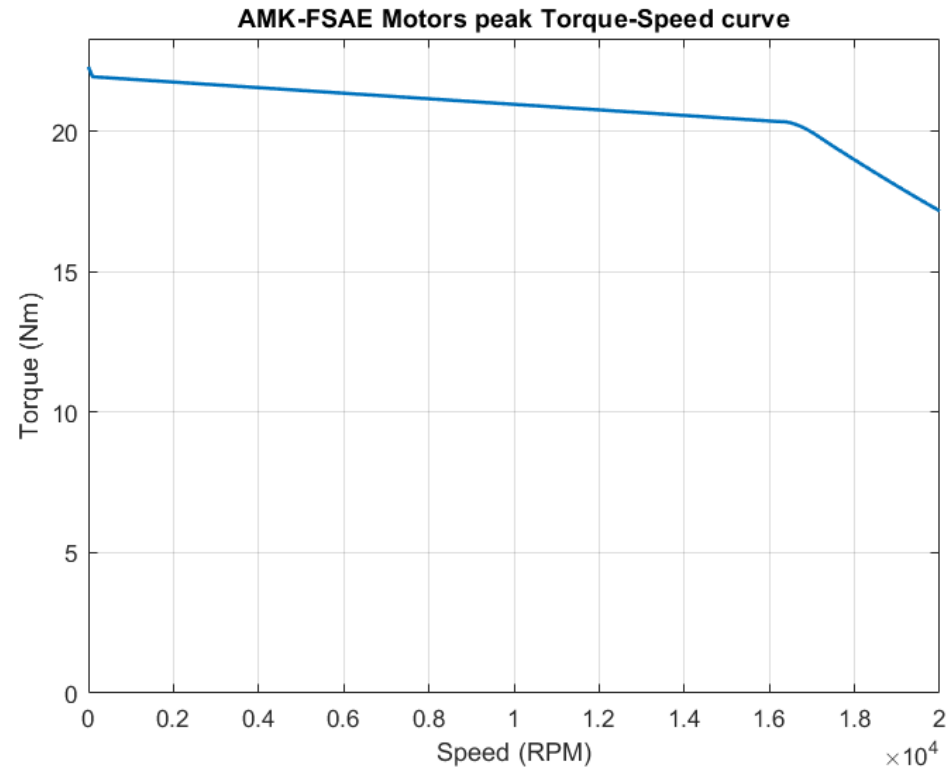
Introduction to Torque Vectoring

- Vehicles that feature independently driven wheels can distribute wheel torque accordingly to assist in car control.



- These plots correspond to motor torques of a 4 wheel independently driven vehicle
- Vehicle is asked to execute an extreme slalom via a pre-calculated steering actuation
- We can see the controller trying to compensate for the vehicles drift from the desired trajectory by redistributing torque

Motor & Tire Choice

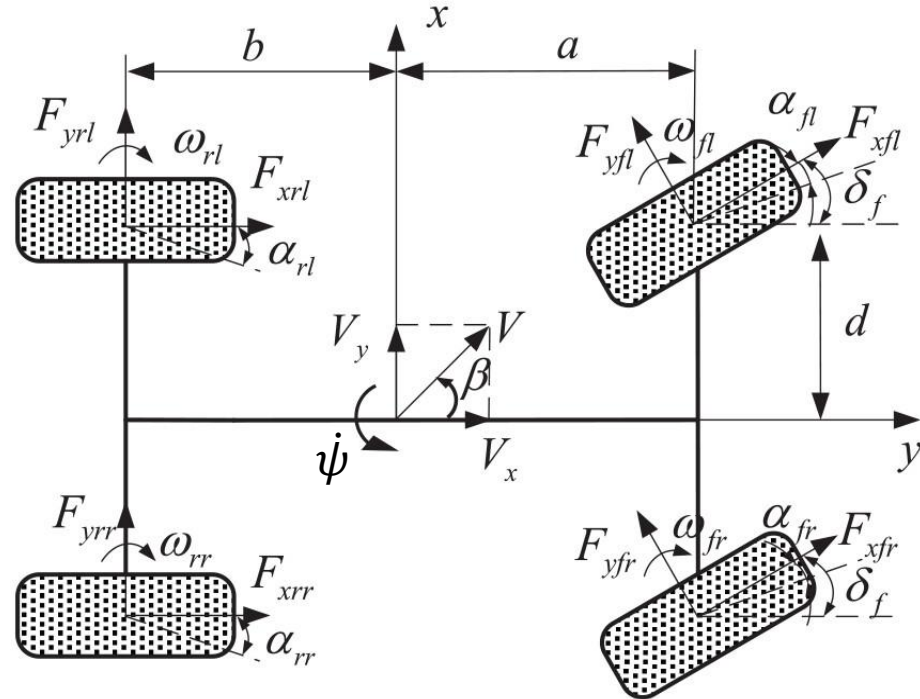


- Motor of choice is the AMK FSAE motor. For simplicity, we are only using the above torque speed curve as a threshold



- Tire Compound of choice is the Hoosier R20, ensuring maximum available grip while keeping weight and inertia to a minimum

Car Planar Dynamics



To simulate vehicle behaviour, we are only considering the planar equations of motion and each wheel's rotation, giving us 7 DoF.

Equations of motion:

$$\dot{v}_x = 1/m ([F_{xfl} + F_{xfr}] \cos \delta + F_{xrl} + F_{xrr} - [F_{yfl} + F_{yfr}] \sin \delta - F_{x_{aero}}) + \psi v_y$$

$$\dot{v}_y = 1/m ([F_{xfl} + F_{xfr}] \sin \delta + F_{yrl} + F_{yrr} + [F_{yfl} + F_{yfr}] \cos \delta) - \psi v_x$$

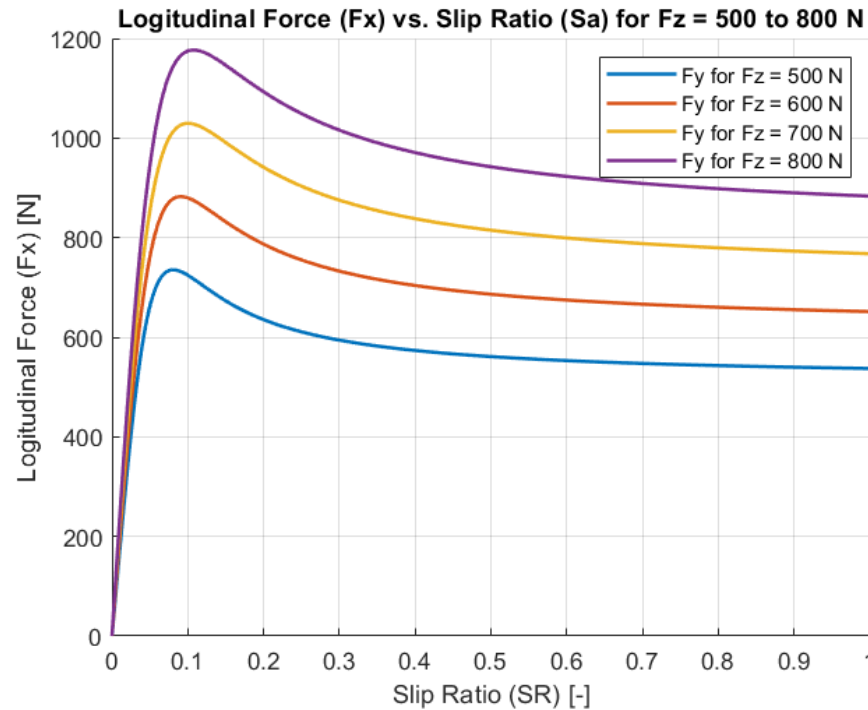
$$\ddot{\psi} = 1/J_z \{ a([F_{xfl} + F_{xfr}] \sin \delta + [F_{yfl} + F_{yfr}] \cos \delta) - b(F_{yrl} + F_{yrr}) + d([-F_{xfl} + F_{xfr}] \cos \delta - [F_{yfl} + F_{yfr}] \sin \delta) + d(-F_{xrl} + F_{xrr}) \}$$

$$\dot{\omega}_{ij} = 1/J_w (T_{wheel} - F_{xij} R - C_{RR} F_{zij})$$

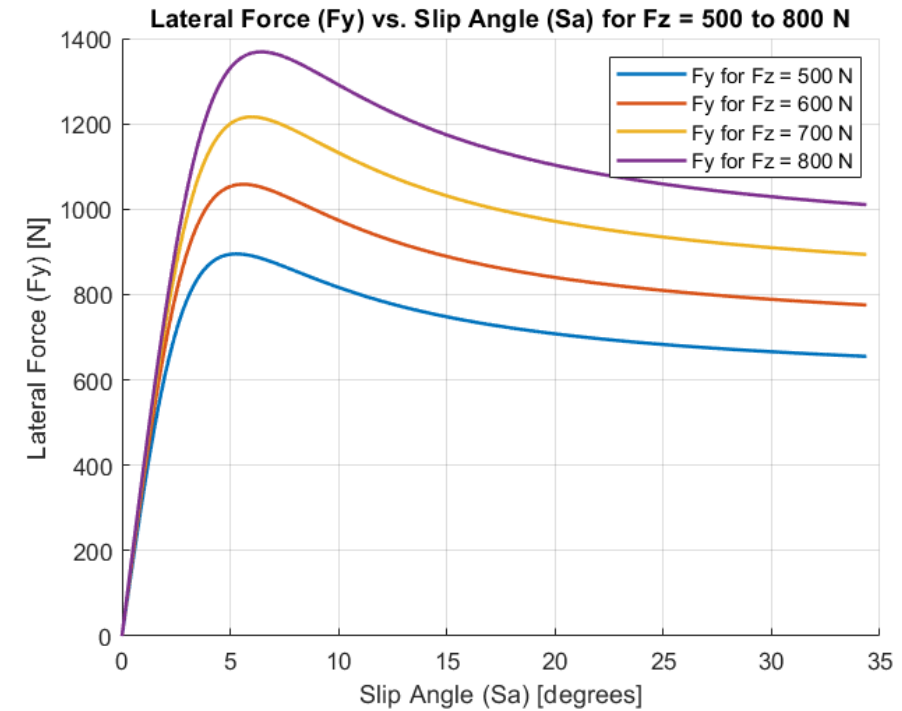
Important note : We consider a drag element that only acts on the x axis, which is not entirely accurate. Ignoring the self aligning torque each wheel produces when the contact patch is deformed was also taken as an assumption to simplify the model.

Tire Forces

- Tire forces are dependant on vertical force, and the longitudinal or lateral deformation of the tire contact patch, known as slip ratio (s_{ij}) and slip angle (a_{ij}) accordingly.



$$s_{ij} = \frac{\omega_{ij}R - (v_x \pm \dot{\psi}d)}{v_x \pm \dot{\psi}d}$$



$$a_{ij} = \text{atan} \left(\frac{v_y \pm \dot{\psi}a}{v_x \pm \dot{\psi}d} \right)$$



Controllers

- We will discuss two different control approaches for the torque vectoring problem. Each one will be presented with their own benefits and drawbacks

State Space Controller :

- + Well established design tools and clear mathematical framework
- + Computational simplicity
- + Ease and straightforward handling of MIMO
- Difficulty of handling heavily non-linear systems
- Implementation and tuning can be complex and very time consuming

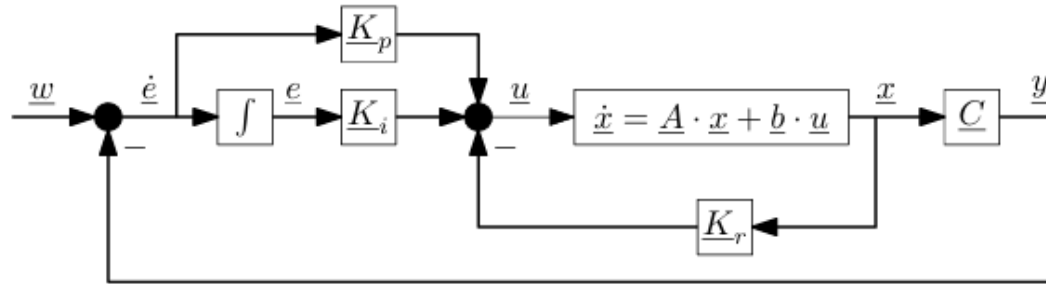
Fuzzy Controller :

- + Straightforward to implement
- + Easy and intuitive tuning
- + Good performance despite nonlinear system dynamics
- Dependence of expert knowledge of the system controlled
- No structured approach for MIMO, needs separate algorithm

State Space Controller

State space controller consists of two parts:

- PI “pre-filtering” to cope with model discrepancies and steady deviation
- LQR for linearized system dynamics



$$\dot{\underline{x}} = \frac{\partial f}{\partial \underline{x}} \underline{x} + \frac{\partial f}{\partial u} u, \quad \underline{x} = \begin{bmatrix} \beta \\ \dot{\psi} \end{bmatrix}, \quad u = [F_{xfl} \ F_{xfr} \ F_{xrl} \ F_{xrr}]^T$$

$$C = [0 \quad 1]$$

- K_r gains are calculated, solving for the gain that minimizes the Ricatti equation of this cost function, where Q and R matrices are tuned to improve performance:

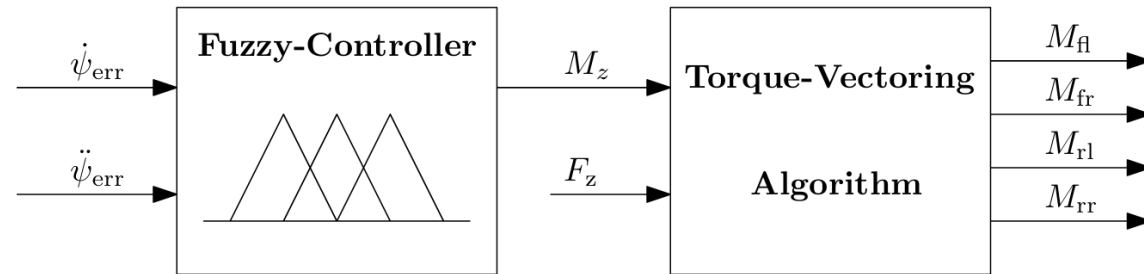
$$J = \int_0^{\infty} \underline{x}(t)^T Q \underline{x}(t) + u^T(t) R u(t) dt$$

- K_i and K_p gains are chosen arbitrarily when tuning
- Linearization is performed around a straight drive

Fuzzy Controller

Fuzzy controller also consists of two parts:

- Fuzzy controller that returns a target M_z to follow a specific ψ
- An algorithm that translates target M_z to wheel torque



M_z	$\ddot{\psi}_{err}$	Error Yaw Rate $\dot{\psi}_{err}$						
		NB	NM	NS	Z	PS	PM	PB
Error Yaw Acc.	NB	NBB	NBB	NBB	NB	NM	NS	Z
	NM	NBB	NBB	NG	NM	NS	Z	PS
	NS	NBB	NB	NM	NS	Z	PS	PM
	Z	NB	NM	NS	Z	PS	PM	PB
	PS	NM	NS	Z	PS	PM	PB	PBB
	PM	NS	Z	PS	PM	PB	PBB	PBB
	PB	Z	PS	PM	PB	PBB	PBB	PBB

- The M_z output of the controller is converted to a total delta F_x between front and rear axis:

$$\Delta F_x = M_z / 2d$$

- ΔF_x calculate is distributed to each wheel based on wheel vertical load F_z :

$$M_{ij} = \pm \frac{\Delta F_x F_{zij} Ratio}{\sum F_{zij}}$$

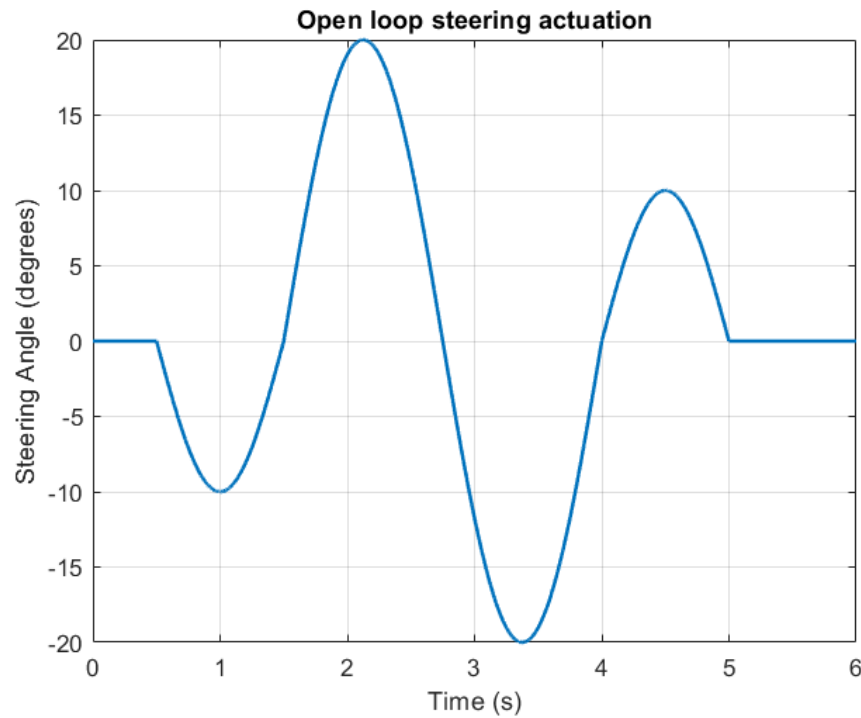
- Error terms are calculated as follows:

$$\dot{\psi}_{err} = \dot{\psi} - \frac{v\delta}{wb}$$

$$\ddot{\psi}_{err} = \ddot{\psi} - \frac{\dot{v}\delta + v\dot{\delta}}{wb}$$

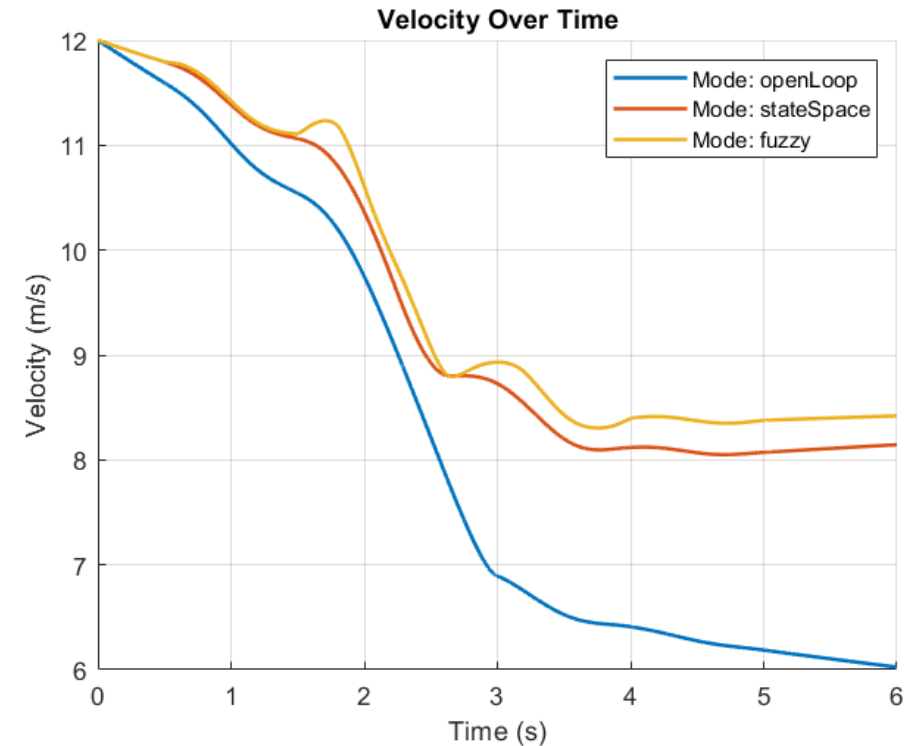
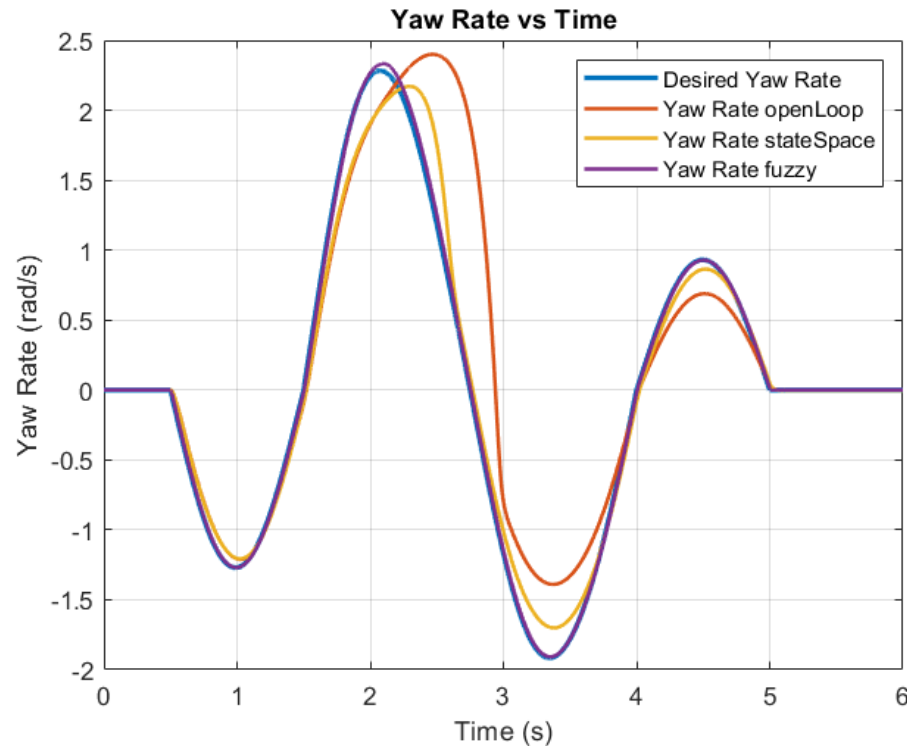
Controller Evaluation

- Evaluation based on open loop steering actuation, inspired by the narrowest possible slalom track layout
- Target is to increase speed taken through the slalom, while minimizing $\dot{\psi}_{err}$.



- Considering actual competition data, this specific combination of sinusoidal functions was selected as it best reflects the most extreme steering sequence observed in a slalom
- The point of choosing the most extreme conditions, is to stress each controller to the maximum

Controller Evaluation



- Overall performance is improved when some form of Torque Vectoring is applied
- Fuzzy controller clearly outperforms state space controller both on $\dot{\psi}_{err}$ reduction and on average velocity maintained through the slalom
- State space controller obviously struggles in the highly nonlinear regions, but performs decently in regions close to linearized dynamics

Conclusions

- A noticeable improvement in performance is observed when Torque Vectoring is used
- Due to the complexity and nonlinearity of the system, a fuzzy controller is considered the preferred approach to handle the torque distribution
- A fuzzy controller should yield better results in real world testing as it does not depend on model accuracy and can easily be tuned to fit the real vehicle on the spot

TV Type	Sum $\dot{\psi}_{err}$	Avg Velocity [m/s]
No TV	0.299	7.88
State Space TV	0.125 (-58.2%)	9.23 (+17.3%)
Fuzzy TV	0.025 (-91.6%)	9.54 (+21.1%)



References

1. Tire model used in this project is the Pacejka Magic Formula 6.2 fitted to TTC data provided by the Vehicle Dynamics department of Prom Racing Formula Student NTUA
2. Jazar, N. (2018) Vehicle Dynamics: Theory and Application. 3rd Edition, Springer-Verlag, New York.
3. B. Jäger, P. Neugebauer, R. Kriesten, N. Parspour and C. Gutenkunst, "Torque-vectoring stability control of a four wheel drive electric vehicle," 2015 IEEE Intelligent Vehicles Symposium (IV), Seoul, Korea (South), 2015, pp. 1018-1023, doi: 10.1109/IVS.2015.7225818.